

STEALTH GROUND-ATTACK AIRCRAFT

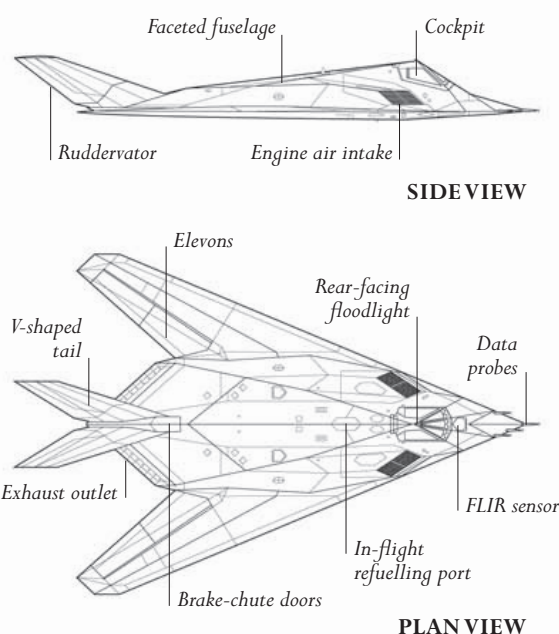
LOCKHEED F-117 NIGHTHAWK

The first operational aircraft designed around stealth technology, the F-117 used faceting – sharply angled surfaces – to reduce its radar signature, enabling it to strike undetected at heavily defended targets.

The F-117 was a direct descendant of Have Blue, an experimental stealth aircraft that Lockheed began developing in 1975. The USAF received its first F-117s in 1982, but the plane's existence remained a secret until 1988. Despite its F (fighter) designation, the Nighthawk was a ground-attack aircraft. It typically carried a pair of 910kg (2,000lb) laser-guided bombs in its internal weapons bay, but had no air-to-air capability. During the Gulf War of 1991, F-117s carried out more than 40 per cent of all strategic air strikes. The aircraft was retired in 2008.

Stealth technology gave the F-117 a radar signature equivalent to that of a small bird. Its angled surfaces scattered incoming radar waves instead of reflecting them back at

their source, and a coating of matt-black radar-absorbent material (RAM) further reduced the signature. To avoid detection the aircraft could not use radar itself, so navigation was by GPS and an inertial guidance system. Efforts to conceal the Nighthawk from heat-seeking missiles included dispensing with afterburners on the engines and cooling the exhaust by channelling it through long ducts lined with heat-absorbent material.



▲ LOCKHEED F-117 NIGHTHAWK

Nearly 20.1m (66ft) long and with a wingspan of 13.2m (43½ft), the F-117 was powered by two General Electric F404 non-afterburning turbofans.

DATA SENSORS



▲ DATA PROBES

Projecting from the tip of the Nighthawk's nose were four short probes that collected data on the aircraft's airspeed and angle of attack.



▲ INFRARED SENSOR

The Forward-Looking Infrared Sensor (FLIR) was used to locate targets. There was also a downward-looking sensor (DLIR) by the front undercarriage.

► IN FLIGHT

Stealth affected flying abilities: the F-117 was limited to subsonic speeds, and its shape made it unstable. Known as the "Wobblin' Goblin", it could be flown only with a computer-controlled fly-by-wire system.

